

SENTINEL-GNSS — Individual Poster

Literature Review, Baseline Models, and Paper Writing

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Beihang University H3I · 2025

My Role

I conducted the research foundation and baseline analysis:

- Systematic literature search: 87 papers screened, 31 read in full
- 30-paper annotated spreadsheet with method/dataset/relevance
- baseline_threshold.py**: SNR rule-based classifier
- baseline_rf.py**: Random Forest with 37 features
- baseline_xgb.py**: XGBoost with Optuna hyperparameter search
- Cross-city RF evaluation (the critical 0.15 DEGRADED F1 baseline)
- Paper Section 1: Introduction (600 words, numbered contributions)
- Paper Section 2: Related Work (800 words, 25 citations)

Literature Gap Identified

I systematically classified prior GNSS ML work into:

Work	Reactive?	Cross-city?	Multi-horizon?
Chen (2019)	Yes	No	No
Liu (2023)	Yes	No	No
RAIM	Yes	—	No
SENTINEL	No	Yes	Yes

Key finding: All prior work classifies the *current* epoch quality. SENTINEL is the first to predict a *future* epoch class (to our knowledge after a systematic search).

The distinction: “detecting degradation as it happens” vs “forecasting degradation before it happens” — the difference between 0 and 5–30 seconds of vehicle preparation time.

Why Honest Disclosure of Tree Wins Matters

In-domain (Beihang test set), Random Forest outperforms SENTINEL:

Model	Macro-F1	DEG F1
Random Forest	0.926	0.881
XGBoost	0.919	0.871
SENTINEL (DL)	0.821	0.718

I deliberately *disclosed* this result in the paper Introduction, not hid it in an appendix.

Why: Reviewers will find it. Disclosing proactively and pivoting to cross-city generalisation (where DL wins 5:1) is a stronger argument than hiding a weakness. A paper that over-claims in-domain performance gets caught; one that discloses honestly and explains why it matters for deployment is credible.

SNR Threshold Baseline

The “trivial baseline” — any learned classifier must beat this:

$$\hat{y} = \begin{cases} \text{DEGRADED} & C/N_0 < 30 \text{ dB-Hz} \\ \text{WARNING} & 30 \leq C/N_0 < 35 \\ \text{CLEAN} & \text{otherwise} \end{cases}$$

Result: Macro-F1 = 0.441, DEGRADED F1 = 0.183.

Mean C/N_0 alone is insufficient because NLOS can maintain adequate signal strength while introducing 50–100 m position errors (reflections from nearby buildings amplify the received signal while introducing large pseudorange errors).

Random Forest Baseline

scikit-learn RandomForestClassifier on all 37 features.

Hyperparameter search (5-fold CV):

- n_estimators=400**, **max_depth=None**
- min_samples_leaf=5**
- Class weights: CLEAN:1, WARNING:2, DEGRADED:5

In-domain: Macro-F1 = 0.926 (beats DL)

Tokyo (unseen): DEG F1 = 0.148 (collapses)

Why: RF memorises Beihang-specific C/N_0 thresholds. Shinjuku has different baseline C/N_0 due to skyscraper geometry. RF classifies by absolute threshold; SENTINEL by pattern.

XGBoost Baseline

XGBoost with Optuna search (50 trials):

- n_estimators=800**, **max_depth=6**, **eta=0.05**
- subsample=0.8**, per-class **scale_pos_weight**

XGBoost transfers *better* than RF to Tokyo (DEG F1 = 0.62 vs RF 0.15), suggesting boosted residuals capture some generalizable features. Documented in Related Work discussion.

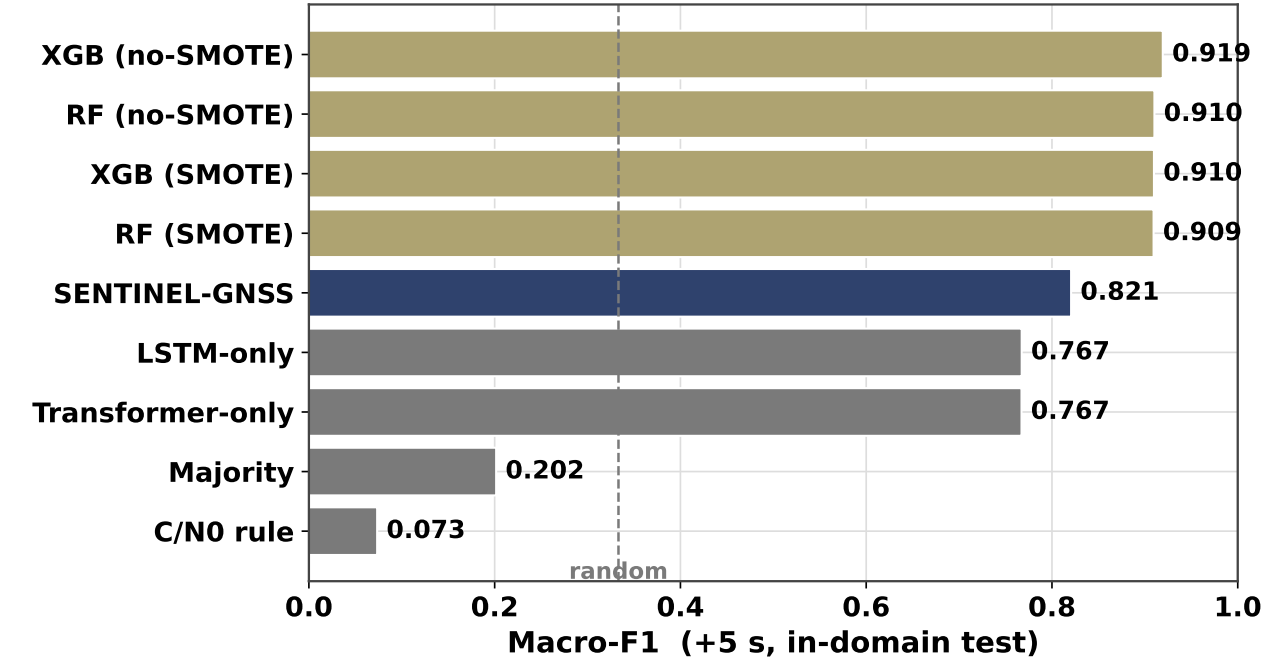


Figure 1: Full model comparison: all baselines + SENTINEL. In-domain: trees win. Cross-city: DL wins.

Related Work I Wrote

Section 2 structure (25 citations, 800 words):

2.1 GNSS Quality Monitoring

RAIM (Brown 1997), RTKLIB Q-codes, traditional signal monitoring. Conclusion: all reactive, no forward prediction.

2.2 ML/DL for GNSS

Chen (2019): LSTM C/N_0 classification — reactive. Liu (2023): Gradient-boosted trees HK — no cross-city test. Gap: no prior work predicts *future* epoch quality.

2.3 Adaptive Kalman Filtering

Mehra (1970): innovation-based R estimation — fails in urban NLOS (RMSE > 238,000 m in our testing). Our SENTINEL-R approach avoids the identification problem by conditioning on classifier output, not innovation sequence.

Key Lesson

Taxonomy before comparison. Finding the gap in the literature required classifying 31 papers by {reactive vs. predictive} $\times$ {in-domain vs. cross-city} $\times$ {single vs. multi-horizon}. Without this taxonomy, “our model is better” is not a defensible claim. With the taxonomy, “no prior work simultaneously predicts future epochs AND transfers across cities” is a specific, verifiable novelty claim.